MSC CYBERSECURITY & DATA SCIENCE · ESAIP ANGERS · GROUP 2

TrustNav

Can a fleet believe its satellite position?

A trust score from 0 to 100 for every vehicle, every second. Explained in plain words. Acted on before a lie reaches dispatch.



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Tutor: Sofiane Hamrioui · Oral examination, 13 November 2026

THE PROBLEM

Everything that moves trusts GNSS blindly

1 500

flights spoofed per day around the Baltic and conflict zones, summer 2024. A daily event for aviation since 2022.



Nantes airport, 2017
A jammer left running in a parked car delayed several flights. Fine: €2 000.



Lyon airport, 2023
A truck driver's jammer, used to escape fleet tracking, disrupted signals until the ANFR traced it.

WHAT EXISTS TODAY

Fleet telematics
shows whatever position it is given

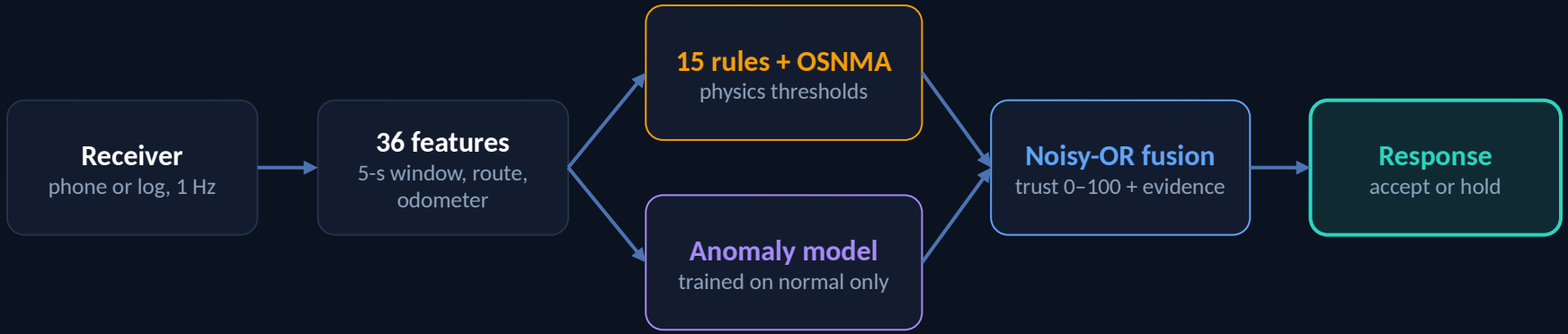
Receiver flags
a state code, no explanation

Fixed probes, anti-jam antennas
protect a site, not a moving van

The gap Nobody tells the operator, per vehicle and per second, whether the position is true, or acts on it.

THE IDEA

A software layer beside the receiver



Scores

0 to 100, per vehicle and drone, every second

Explains

in plain words, with values: “Fix moved 7 998 m in 1 s (limit 250 m)”

Responds

refuses the fix and tracks the vehicle by its odometer along its route

DETECTION

Three witnesses, one verdict

Rules

15 physical checks with named thresholds

CATCHES a van cannot jump 250 m in a second

MISSES slow attacks never cross a threshold

60 m/s · 8 m/s² · 6 dB-Hz C/N0 shift · PDOP > 5

Anomaly model

Isolation Forest + tail detector, trained on normal driving only

CATCHES a fix 90 m off the road with clean signals

MISSES noisier; must be calibrated on the receiver

risk = 25 · (−log10 p): 50 is a 1-in-100 event

Galileo OSNMA

satellites sign their navigation data (operational since July 2025)

CATCHES forged data: the signature fails

MISSES relay attacks keep it valid; GPS not covered

one witness among three, not a defence

Fused by noisy-OR, not averaged

risk = 100 · [1 − (1 − 0.6 · p rules)(1 − 0.8 · p model)] trust = 100 − risk

MODEL SAYS 90
RULES SAY 20

55 average hides it

75 noisy-OR alerts

The response layer: the lie stops here

1

ACCEPT

trust at least 50 and no critical rule: the fix passes through unchanged.

2

HOLD

the fix is refused. The position handed on is dead-reckoned from the last trusted fix with the odometer, along the planned route. A spoofer can touch neither.

3

RELEASE

after 5 clean seconds and a fix within 60 m + 3 % of the distance travelled. After 60 s the bound is ten times looser; a fix still kilometres away stays refused.

CONTINUE VERIFY ISOLATE



The console during the teleport: the van's real path (left), the reported position 8 km away (right), and the dashed lie between them.



What the operator reads

Score 15, Untrusted. Rules 45, model 100, combined risk 85.

"The receiver places the van 8.0 km from the road it is actually on… TrustNav has stopped passing this position on."

EVALUATION

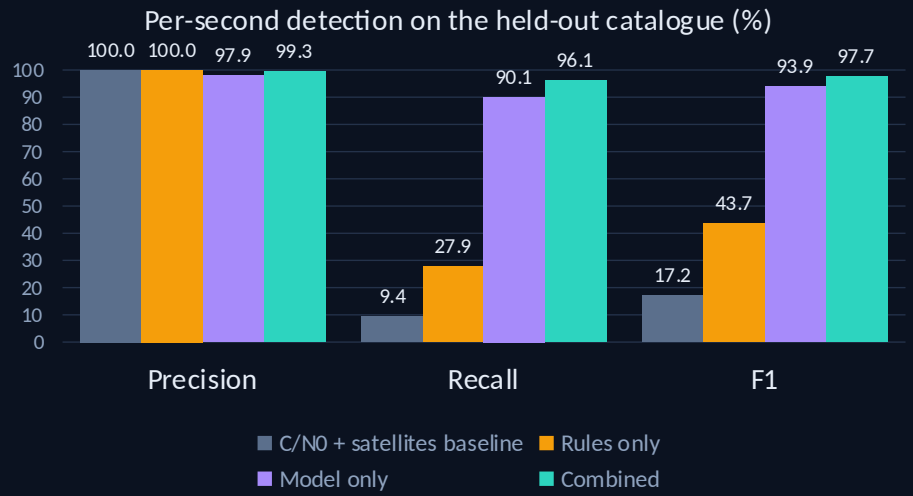
Tested on routes it has never seen

99.3 %
PRECISION

96.1 %
RECALL

0.44 %
FALSE ALARMS

1 s
MEDIAN DELAY



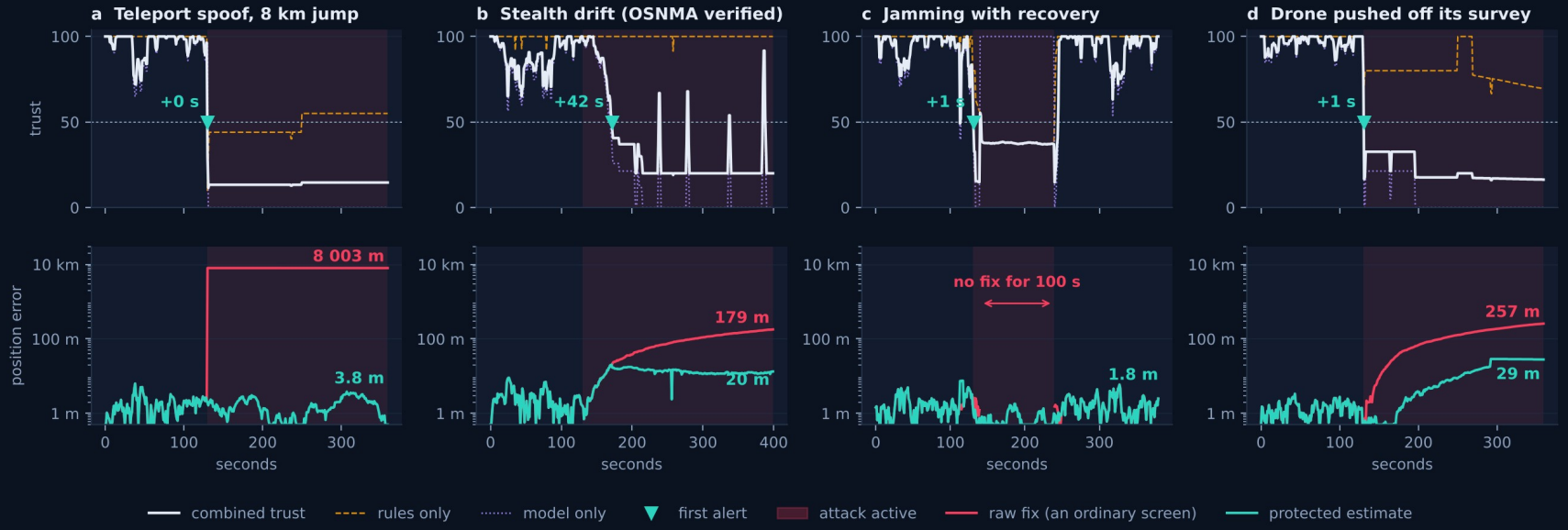
10 scenarios · 3 800 seconds
1 550 under attack, 7 attack types, real Angers streets

One role per route
train, validate or test: never shared

95 % intervals over scenarios
precision 97.9–100 · recall 89.4–99.6

FOUR HELD-OUT ATTACKS

What happens, second by second



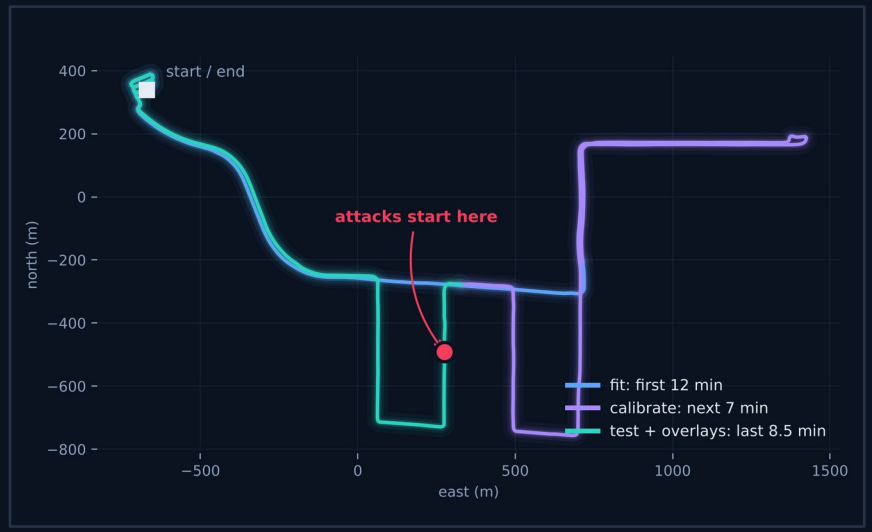
7 / 7
attack episodes detected

8 km → 29 m
worst position handed on

100 s → 0 s
blank screen under jamming

42 s
stealth drift caught, 12 m off the road

Then we tried a real receiver



Public recording: u-blox ZED-F9K with a NovAtel SPAN reference, Mountain View, 27.6 min (Google gps-measurement-tools, Apache-2.0). The reference track serves as route and odometer, so protected errors are idealised.

0 — **false alarms from the rules**
in the whole drive (minimum trust 90). The classic signal baseline raised 86 s.

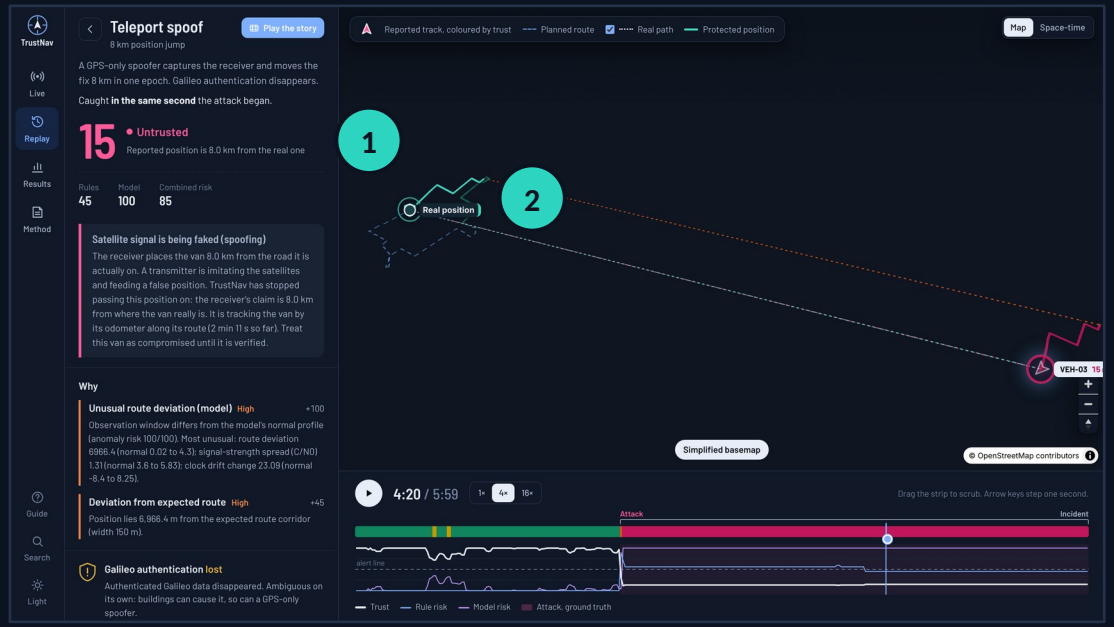
89 % — **of seconds alerted by the simulator-trained model**
so it must be re-calibrated on the receiver in use. Re-calibrated: no alert in the last 8.5 min.

0 · 35 · 1 s — **to catch overlaid teleport, drift, jamming**
protected estimate within 0.8, 24 and 0.8 m of the reference.

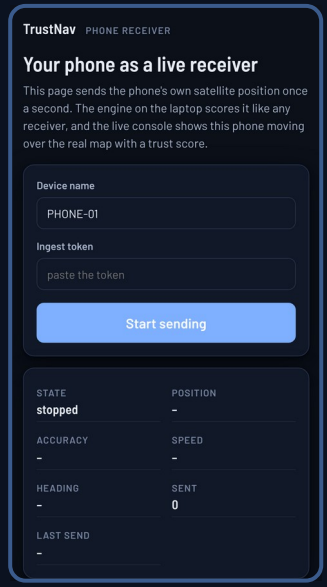
Real data found two weaknesses. Both fixed.
Tail scores now in log space · the 60-s release is bounded

LIVE DEMO

The console, and a phone as a receiver



3



1 score, verdict, evidence **2** real path vs the reported lie **3** your phone, scored live, with an attack switch

trustnav.vercel.app · open source · Python engine, 32 tests · Supabase, row-level security · Next.js console, 109 end-to-end checks

HONEST CLAIMS

What we claim, what we don't, what's next

Limits

- one simulator: the numbers are an upper bound
- route checks need an assigned route
- the response is as good as its odometer
- OSNMA simulated; no multi-antenna evidence

Safety and ethics

- nothing is transmitted: attacks are software
- future hardware is receive-only
- GDPR: purpose, 3-day retention, transparency
- a research indicator, not certified integrity

Next

- hours of real drives in Angers
- replay Jammertest 2023 and TEXBAT captures
- a second receiver against synchronous spoofing
- real OSNMA from raw Galileo pages

A per-second, explained trust score, and a response that keeps a lie from the systems that act on it.

Questions?

trustnav.vercel.app · github.com/domail2karandeshmukh-cyber/trustnav